Piyush Singh

Data Scientist

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Summary

Data Scientist with 2 years of experience developing LLM-powered and machine learning systems for NLP, computer vision, and analytics. Skilled in Python, PyTorch, LangChain, and cloud platforms (AWS, GCP, Docker). Passionate about building ethical and scalable AI solutions that bridge research with real-world applications.

Experience

AI Consultant — Feb. 2025 – May 2025
DeHaat — *Remote, IN*

- Consulted DeHaat to build a multilingual intent and entity recognition framework integrated into their chatbot ecosystem, including dataset development, model performance documentation, and scalable architecture planning for future services.

Data Scientist — Jan. 2024 – Aug. 2024
DeHaat — *Gurgaon, IN*

- Built an LLM-powered agronomist assistant tool for farmers, funded by the Gates Foundation, using LangChain, VectorDB, Hugging Face, and ReactJS for a functional and testable frontend-backend system.
- Developed Flask APIs and implemented RLHF training pipelines, including DPO fine-tuning and function calling, to enhance the capabilities of an open-source chat-based model.
- Automated data collection with Selenium and BeautifulSoup to generate a domain-specific VectorDB, while actively integrating state-of-the-art NLP techniques through ongoing research.

Junior Data Scientist — June 2023 – Jan. 2024
Nuvoretail — *Delhi, IN*

- Developed and deployed transformer-based and random forest ML models to optimize search advertising campaigns and quantify Amazon market share, achieving a 73% R^2 score and driving strategic insights.
- Built a custom optimization algorithm for dynamic advertising budget allocation, improving resource efficiency across campaigns.
- Engineered a web scraping pipeline using Selenium to extract Amazon product data and generate high-quality datasets for machine learning model training.

Senior Project Assistant — April 2022 – Aug. 2022
Indian Institute of Technology — *Delhi, IN*

- Led an ICMR-funded research project to design and develop a nerve monitoring system prototype, programming microcontrollers and integrating communication protocols.
- Gathered and analyzed skin impedance data, optimized system performance, and documented progress in reports and presentations.

Internships

Machine Learning Intern — Sep. 2021 – Nov. 2021
Knowledge Solutions India — *Remote, IN*

- Developed SQL scripts to extract and manipulate data from relational databases, such as MySQL.
- Gained hands-on experience in machine learning techniques such as regression, clustering, and classifications techniques.
- Conducted data preprocessing, feature engineering, and exploratory data analysis to gain insights into the dataset

Project Management Intern — June 2019 – July 2019
Larsen & Toubro — *Mumbai, IN*

- Maintained bill of material using Advanced Excel, ensuring accurate tracking of project resources and materials.
- Conducted research and created a comprehensive report on 'Hydraulic Systems' as part of academic coursework, outlining the design and functioning of hydraulic systems.

Education

The University of Manchester Manchester, UK
MSc in Data Science (Social Analytics) *Sep. 2024 – Sep. 2025*

- Modules: Statistics and Machine Learning, Topological Data Analysis, Cognitive Robotics and Computer Vision.
- **Bias Detection in LLMs:** Developed a multitask DeBERTa-based model to detect political, gender, and immigration bias in LLM outputs, introducing a novel metric for bias assessment. Evaluated and compared bias levels across multiple models to promote fairness and transparency in AI systems [Test Model].
- **Social Listening Platform (Vulse Collaboration):** Built a social listening and advisory web application that collects real-time data from Twitter (X), Facebook, and web sources via official APIs. Integrated LangChain-based LLM pipelines for competitor analysis and marketing recommendations. Developed with Next.js, Node.js, and MongoDB, featuring secure authentication, API documentation, and scalable architecture for enterprise deployment.

Imarticus Learning Delhi, IN
Post-Graduate Program in Data Analytics and Machine Learning *Aug 2022 – Apr 2023*

- Covered statistical foundations, Python and R for Data Science, SQL programming, and data visualization using Tableau and Power BI.
- **Neural Machine Translation using Transformer Architecture:**
  - Built a Transformer-based neural machine translation model for English to Hinglish, using TensorFlow and Keras with custom encoder-decoder architecture and positional embedding layers.
  - Preprocessed and vectorized text data using a custom CharBPE tokenizer via Hugging Face, creating efficient input-output pairs for training.
  - Trained and evaluated the model, optimized hyperparameters, and validated performance through real-world translation examples, documenting the full pipeline in a detailed presentation.
- **Stroke Prediction Model using PySpark on Hadoop Cluster in GCP:**
  - Developed and deployed a PySpark-based Random Forest model for stroke prediction on a GCP-hosted Hadoop cluster, achieving 81% accuracy.
  - Performed comprehensive EDA and preprocessing, including data cleansing, normalization, and feature engineering using PySpark ML libraries.
  - Managed GCP Hadoop cluster infrastructure, ensuring scalable computation, and delivered a detailed presentation highlighting methodologies, performance metrics, and insights.

University of Mumbai Mumbai, India
Bachelor of Engineering in Mechanical Engineering *July 2017 – Aug. 2021*

- Relevant Courses: Applied Mathematics, Applied Physics, Mechatronics, Operations Research, Industrial Electronics, Computer Programming in C and Database Management Systems.
- Dissertation — Quadruped Robot: *Designed and built a 12-DOF quadruped robot using SolidWorks and 3D printing; implemented inverse kinematics in C and Python for wireless control. Integrated reinforcement learning with ultrasonic sensors for autonomous obstacle avoidance. Project Links: PSybeR-Cat, Spider*

Publications

A Topology of Educational Outcome Inequality 2025
Benites, A., Hara, K., Singh, P., Zubier, B., & Rudkin, S. *SSRN ID: 5401815*

- Published on SSRN exploring topological approaches to understanding inequality in educational outcomes across social and economic dimensions.

Hackathons

AI Encode London Hackathon London, UK
1st Prize – StarkWare Bounty *Apr. 2025*

- Built a blockchain-based open-world game integrating Generative AI and crypto land trading. Recognized for innovation in combining scalable blockchain infrastructure with immersive AI-driven gameplay.

Projects

Tool Integration Engine (TIE) | *Python, LLMs, Knowledge Graphs, PyTorch Geometric* October 2025 – Present

- Designed and developed a modular engine that enables Large Language Models to dynamically discover and invoke external tools using graph-based reasoning. Tools, parameters, and domains are represented as nodes in a knowledge graph, allowing structured traversal for context-aware API selection. Supports plug-and-play registration for enterprise systems and integrates GNN-based models for learning optimal tool invocation paths.
- Implemented multi-relational graphs in NetworkX and trained GNNs using PyTorch Geometric for tool prediction. Future versions will incorporate reinforcement learning for dynamic routing and multi-step task orchestration.

MIRA (AI Operating System) | *Python, LLMs, RAG, ReactJS* September 2025

Personal Research Project

- Engineered a Linux-based AI Operating System capable of autonomously managing system operations through natural language commands.
- Enabled file system control, OS health monitoring, and task automation, allowing MIRA to manage files, check resource usage, and execute operations on demand.
- Integrated Google Calendar, Gmail, and Notes using OAuth 2.0 for context-aware scheduling, email handling, and note summarization.
- Built a real-time frontend with Vite + React, providing a conversational terminal UI to visualize system activity and AI responses.
- Developed a modular backend with FastAPI, LLM orchestration, and a secure Docker-based execution layer featuring guardrails and audit logging.
- Implemented retrieval-augmented generation (RAG) over Linux command documentation and personal data, enabling intelligent reasoning and contextual decision-making.
- Designed for scalability, privacy, and extensibility, supporting local or dedicated server deployment for a fully autonomous AI OS experience.

ZeroMQ Communication Patterns (Educational Project) | *Python, ZeroMQ* Jun. 2023

- Developed an educational project to teach the five core ZeroMQ messaging patterns (REQ/REP, PUB/SUB, PUSH/PULL, ROUTER/DEALER, PAIR) using Python (`pyzmq`).
- Authored step-by-step, runnable examples demonstrating asynchronous and distributed communication across microservices and network nodes.
- Focused on simplifying complex concepts like socket management, concurrency, and non-blocking I/O for learners and developers.
- Designed reusable templates and clear documentation to serve as a learning resource for distributed systems and IPC fundamentals.
- Published publicly to help others understand ZeroMQ's core communication models. GitHub Link

LazyForecast: A Python Library | *Python, GitHub Actions, TensorFlow, Machine Learning* Apr. 2023

- Developed and published **LazyForecast** on PyPI — a Python library for lazy time-series forecasting, supporting models such as ARIMA, MLP, and LSTM. It automates data preprocessing, model training, and evaluation with minimal code.
- Integrated auto-ARIMA via `pmdarima` for automatic hyperparameter optimization, and added visualization utilities for forecast diagnostics and interpretability.
- Set up GitHub Actions for CI/CD, automating PyPI releases and ensuring robust package deployment. PyPI Link, GitHub Repo

Achievements

- Delivered Guest Lecture at the Ministry of Defence (2023): Presented on AI concepts and conducted a hands-on training session for MoD employees.
- Cleared GATE (2024): Qualified the Graduate Aptitude Test in Engineering for the Data Science and Artificial Intelligence specialization.
- Scored 7.5 Bands in IELTS (Mar 2024): Demonstrated strong English proficiency across listening, reading, writing, and speaking.
- Led Industrial Visit to Indian Ordnance Factory (2019): Organized and coordinated a successful visit for the Mechanical Engineering Department.
- Bronze Medal – National Science Olympiad (NSO) (Dec 2014): Recognized for academic excellence at the national level in science.

Certificates

- Azure AI Fundamentals – AI900 by Microsoft
- Machine Learning with AI using Python by Certiport
- Microsoft AI Classroom Series by Microsoft
- Introduction to Big Data by UC San Diego via Coursera
- Machine Learning: Regression by University of Washington via Coursera
- Python Data Structures by University of Michigan via Coursera
- AI for Everyone by Deeplearning.ai via Coursera
- A1 – German (Deutsch) Language via Goethe Institut Indien

Workshops

- Attended a 2-day workshop on "Machine Learning" at IIT Bombay (24–25 Aug 2019).

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, R
Frameworks: React, Node.js, Flask, Material-UI, FastAPI, DJango
Databases: MS SQL, MySQL, Hive, GraphDB
Developer Tools: Git, Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), VS Code, Visual Studio, GitHub Actions, Linux, Sagemaker, Bedrock, Hadoop
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Sci-kit Learn, TensorFlow, Keras, OpenCV, PySpark, BeautifulSoup, Selenium, YOLO, PyTorch, LangChain, LangGraph
Other: Advance Excel, Tableau, LLM, RAG, , Agentic-AI, Reinforcement Learning, GraphDB, Graph Neural Network (GNN), ZeroMQ (pyzmq)

Soft Skills

Business Acumen **Problem-Solving** **Collaboration** **Adaptability** **Communication**

Expertise

Machine Learning
Deep Learning
Statistical Analysis
Data Visualization
Data Preparation
Data Structures & Algorithms
Data Exploration
Natural Language Processing
Computer Vision